Do Now · retrieval

Lesson 063 · Thinking Logically: Conditions and Decisions

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does a Boolean expression evaluate to?

Q2

What is a decision boundary?

Q3

Why is condition order important in else-if?

Q4

Simplify (A AND B) OR (A AND NOT B), showing the Boolean laws used.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify — Identify decisions in the solution and write precise Boolean conditions, including valid-data boundaries.

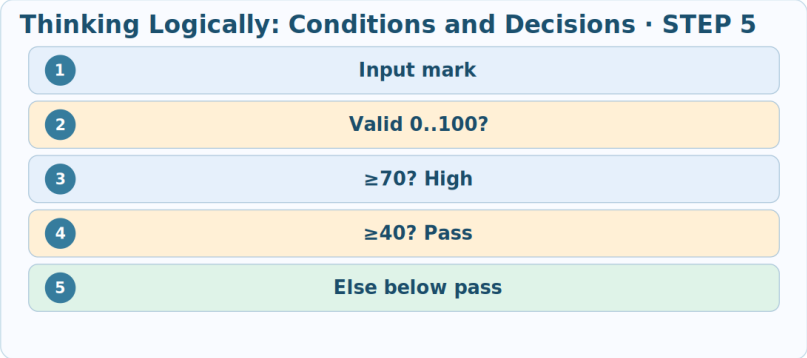
Core idea 2

Model — Order conditions so cases are mutually understood: mark>=70 must be considered before mark>=40.

Core idea 3

Evaluate — Use truth tables or decision tables to test normal, erroneous and boundary input values.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

predicate — Boolean condition.

selection — Branch based on a condition.

boundary test — Input near comparison threshold.

short-circuit — Skipping redundant Boolean evaluation.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

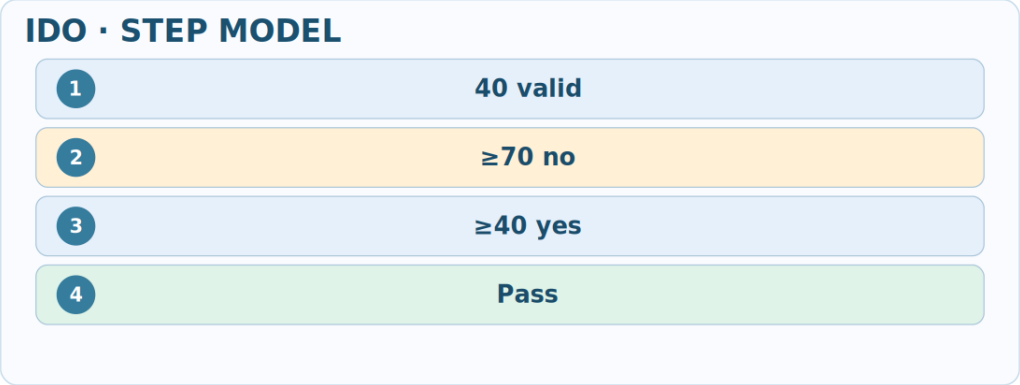
Problem

Trace Grade(40) using the conditions shown.

Follow the worked method

1. Explain the decision — Check comparisons in order; stop after first matching return.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the decision
Without range check, 105 would incorrectly be classified High.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace Grade for marks 39,40,69,70 and give output at each. [4]

Question 2

Write a Boolean rule for a valid age 11–18 inclusive, and give two boundary tests. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A program checks if(mark>=40) before else-if(mark>=70), then outputs Pass for 85. Explain bug and repair. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

With else-if, a later condition always runs even if an earlier branch matches.

Use key vocabulary from memory

1. Explain predicate in your own words.

2. Explain selection in your own words.

3. Explain boundary test in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.